Further Mathematics GCE A Level 2022 - Paper 1

Duration: 1h30 | Cameroon GCE Board

GENERAL CERTIFICATE OF EDUCATION BOARD, General Certificate of Education Examination. 0775 FURTHER MATHEMATICS 1, ADVANCED LEVEL, JUNE 2022. Mobile phones are NOT allowed in the examination room. Multiple choice question paper. Duration: one and a half hours. Use a soft HB pencil throughout the examination. Answer ALL the 50 questions in this examination. All questions carry equal marks. Calculators are allowed. Each question has FOUR suggested answers: A, B, C and D. Decide which answer is correct, find the number of the question on the answer sheet and draw a horizontal line across the letter to join the square bracket for the answer you have chosen. Mark only one answer for each question (if more than one answer is marked, the question scores zero). Avoid spending too much time on any one question: if a question is difficult, move on to the next and come back to it later. Do all rough work in the booklet, using the blank spaces in the question booklet.

---

1. The complementary function of the differential equation $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} = 10x + 1$, where $P, Q \in \mathbb{R}$, is

a) $y = P + Qe^{3x}$
b) $y = Pe^{-3x} + Q$
c) $y = (P + Qx)e^{-3x}$
d) $y = P\cos 3x - Q\sin 3x$

2. $\frac{x+4}{(x^2-1)(x+1)}$, $x \neq \pm 1$, can be expressed in partial fractions, where P, Q and R are constants, as

Options as printed: A) $\frac{P}{x-1} + \frac{Q}{x+1} + \frac{R^2}{(x+1)^2}$ B) $\frac{Px+Q}{x^2-1} + \frac{R}{x+1}$ C) $\frac{P}{x-1} + \frac{Q}{(x-1)^2} + \frac{R}{x+1}$ D) $\frac{P}{x-1} + \frac{Q}{x+1} + \frac{Rx}{(x+1)^2}$

3. The function $f(x) = \frac{(x-3)(x+1)}{x(x-3)(x+3)}$ has a removable discontinuity at $x =$

a) 0
b) 1
c) 2
d) 3

4. $\lim_{x\to\infty} \frac{\cos x}{x} =$

a) ∞
b) 0
c) 1
d) $-\infty$

5. Which one of the polar curves is symmetrical about the line $\theta = 0$ only?

a) $r = 1 + \cos\theta$
b) $r = 1 + \sin\theta$
c) $r = \sin 2\theta$
d) $r = \cos 2\theta$

6. If the motion modelled by $\dfrac{d^2x}{dt^2} + 2k\dfrac{dx}{dt} + n^2x = 0$ is a damped harmonic motion, then

a) \$k b) $k \leq n$
c) \$0 d) $0 \leq k \leq n$

7. Given that $\mathbf{p}$ and $\mathbf{q}$ are two non-zero vectors such that $\mathbf{p} - \alpha\mathbf{q} = \mathbf{0}$, where α is a real constant. Which one of the following is true?

a) $\mathbf{p} \cdot \mathbf{q} = 0$
b) $\mathbf{p} \cdot \mathbf{q} = \alpha$
c) $\mathbf{p} \times \mathbf{q} = \mathbf{0}$
d) $|\mathbf{p} \times \mathbf{q}| = \alpha$

8. The structure $(\mathbb{N}, +)$ is NOT a group because $(\mathbb{N}, +)$

a) is not closed
b) is associative
c) does not have an identity element
d) does not have inverse elements

9. If $2x \equiv 3 \pmod 7$, then $x \equiv$

a) 2
b) 3
c) 4
d) 5

10. Given that $\dfrac{dy}{dx} = x - \dfrac{y}{x}$ and $y = 0$ when $x = 1$, then using the approximation $y_{n+1} \cong y_n + hy'_n$, $h = 0.1$, the value of y when $x = 1.1$ is

a) 0.1
b) 0.01
c) 0
d) 0.2

11. A, B, C are the position vectors of the vertices of a triangle. The area of triangle ABC is

Options as printed: A) $\left|\frac{1}{2}\overrightarrow{AB} \times \overrightarrow{AC}\right|$ B) $\left|\frac{1}{2}\overrightarrow{AB} \times \overrightarrow{BC}\right|$ C) $\left|\frac{1}{2}\overrightarrow{AC} \times \overrightarrow{BC}\right|$ D) $\left|\frac{1}{2}\overrightarrow{AB} \times \overrightarrow{CB}\right|$

12. Given two statements $p:\ \sinh x = \sinh y$ and $q:\ x = y$. The contrapositive of $p \Rightarrow q$ is

a) If $x \neq y$, then $\sinh x \neq \sinh y$
b) If $x = y$, then $\sinh x \neq \sinh y$

c) If $x = y$, then $\sinh x = \sinh y$
d) If $x \neq y$, then $\sinh x = \sinh y$

13. If $\sinh x = \dfrac{1}{2}$, then $\cosh 2x =$

a) $\dfrac{1}{2}$
b) $\dfrac{3}{2}$
c) 1
d) $\dfrac{\sqrt{3}}{2}$

14. Given that $|z| = 3$, the maximum value of $|z - 2i|$ is

a) 1
b) 2
c) 3
d) 5

15. The Taylor series expansion of the function $f(x) = \ln(3 - 4x)$ is valid for

a) $\left]-\frac{3}{4}, \frac{3}{4}\right]$
b) $\left[-\frac{3}{4}, \frac{3}{4}\right]$
c) $\left]-\frac{3}{4}, \frac{3}{4}\right[$
d) $\left[-\frac{3}{4}, \frac{3}{4}\right[$

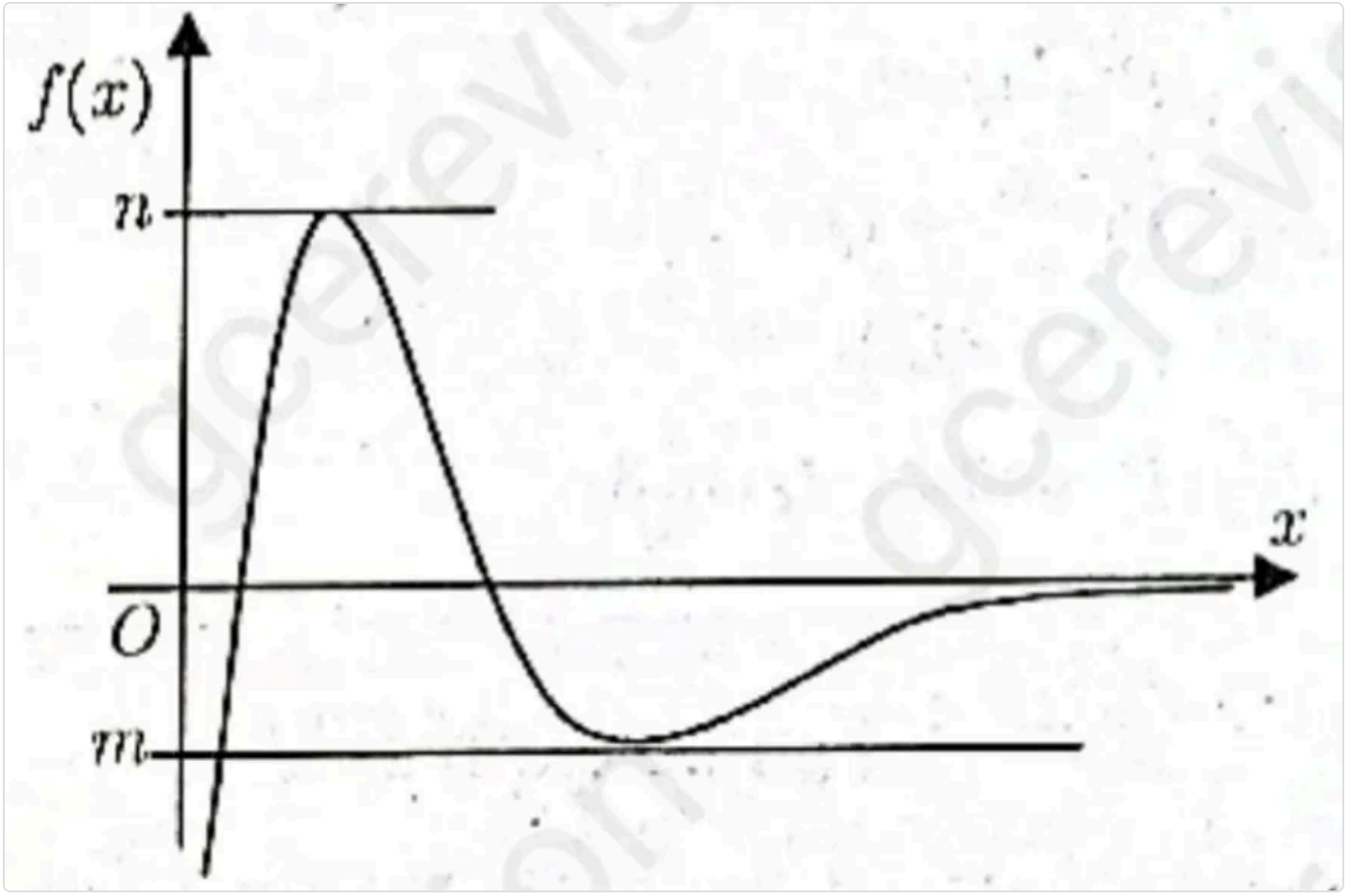
16. If $u_1 = \dfrac{11}{2}$, $u_{n+1} = \lambda u_n$, $n \geq 1$, then u_n can be expressed as

a) $\left(\dfrac{11}{2}\right)\lambda^n$
b) $\left(\dfrac{11}{2}\right)^n \lambda$
c) $\left(\dfrac{11}{2}\right)\lambda^{n-1}$
d) $\left(\dfrac{11}{2}\right)^{n-1} \lambda$

17. For $x > 1$, $\displaystyle\int \frac{1}{x \ln x}\, dx =$

a) $\ln x + k$
b) $\ln(\ln x) + k$
c) $\ln^2 x + k$
d) $2 \ln x + k$

18.



From the graph of f, the equation $f(x) = a, a \in \mathbb{R}$, has only one root for

a) $a > n$
b) $a > m$
c) $n d) $a

19. The root mean square value of $\sinh x$, for $1 \leq x \leq 3$, is

a) $\left[\frac{1}{2}\int \sinh^2 x\, dx\right]^{\frac{1}{2}}$
b) $\frac{1}{2}\left[\int_1^3 \sinh^2 x\, dx\right]^{\frac{1}{2}}$
c) $\left[\frac{1}{2}\int_1^3 \sinh^2 x\, dx\right]^{\frac{1}{2}}$
d) $\frac{1}{3}\left[\int_1^3 \sinh^2 x\, dx\right]^{\frac{1}{2}}$

20. $P(X = x) = e^{-kx}, x \geq 0$. The mean of X is

a) k^2
b) $\frac{1}{k^2}$
c) $-k^2$
d) $-\frac{1}{k^2}$

21. The domain of the function $g(x) = \frac{x+1}{\sqrt{x^2-9}}$ is

a) $\{x \in \mathbb{R} : x < -3\} \cup \{x \in \mathbb{R} : x > 3\}$
b) $\{x \in \mathbb{R} : -3 \leq x \leq 3\}$
c) $\{x \in \mathbb{R} : x \leq -3\} \cup \{x \in \mathbb{R} : x \geq 3\}$
d) $\{x\in\mathbb{R}:-3\le x<-1\}\cup\{x\in\mathbb{R}:-1

22. Given the sequence (u_n) where $u_0 = 3$, $u_{n+1} = \dfrac{2}{3}u_n$, then $\lim\limits_{n\to\infty} u_n =$

a) 0
b) 1
c) $\dfrac{2}{3}$
d) 3

23. The image of the line $y = 2x$ under the transformation matrix $\begin{pmatrix} 1 & 0 \\ 2 & -3 \end{pmatrix}$ is the line

a) $y = x$
b) $y = 2x$
c) $y = -4x$
d) $y = 1 - x$

24. $\displaystyle\int_0^1 \frac{1}{9 - x^2}\,dx =$

Options as printed: A) $\sinh^{-1}\left(\frac{1}{3}\right)$ B) $\cosh^{-1}\left(\frac{1}{3}\right)$ C) $\tanh^{-1}\left(\frac{1}{3}\right)$ D) $\cosh^{-1}(3)$

25. The work done when the point of application of the force $\mathbf{F}$ is displaced from $\mathbf{r}_2$ to $\mathbf{r}_1$ is

a) $(\mathbf{r}_2 - \mathbf{r}_1) \times \mathbf{F}$
b) $(\mathbf{r}_2 - \mathbf{r}_1) \cdot \mathbf{F}$
c) $(\mathbf{r}_1 - \mathbf{r}_2) \times \mathbf{F}$
d) $(\mathbf{r}_1 - \mathbf{r}_2) \cdot \mathbf{F}$

26. A smooth sphere is moving with velocity $(-3\mathbf{i} + \mathbf{j})\ \mathrm{m\,s^{-1}}$ when it hits a smooth wall parallel to the vector $\mathbf{j}$. If the coefficient of restitution between the sphere and the wall is $\dfrac{1}{3}$, then the velocity of the sphere after hitting the wall is

a) $\mathbf{i} + \mathbf{j}$
b) $\mathbf{i} - \mathbf{j}$
c) $-\mathbf{i} + \mathbf{j}$
d) $-\mathbf{i} - \mathbf{j}$

27. Given that $z, w \in \mathbb{C}$, then the transformation $w = -z$ is

a) a rotation $\dfrac{\pi}{2}$ clockwise
b) a rotation $\dfrac{\pi}{2}$ anticlockwise
c) a rotation π clockwise
d) a rotation $\dfrac{-\pi}{2}$ anticlockwise

28. If $25\cosh x + 7\sinh x \equiv R\sinh(x + \lambda)$, then the value of R is

Options as printed: A) $\sqrt{674}$ B) 24 C) 18 D) 16

29. The moment of inertia of a rigid body of mass $2M$ about a given axis is $\frac{Ma^2}{8}$. The radius of gyration about the axis is

a) $\frac{a}{\sqrt{8}}$
b) $\frac{a}{4}$
c) $\frac{d}{2}$
d) $\frac{a^2}{4}$

30. The probability density function of a continuous random variable X is given by $f(x) = \begin{cases} \frac{k}{2}(x^2-1), & -1 \le x \le 1 \\ 0, & \text{elsewhere} \end{cases}$. The value of k is

Options as printed: A) $-\frac{8}{3}$ B) -2 C) $-\frac{3}{8}$ D) $-\frac{4}{3}$

31. The number, n, of bees in a large apiary grows at a rate directly proportional to the number of bees present at time t. The differential equation that models the number of bees in this apiary, for $k > 0$, is

a) $\frac{dn}{dt} = kn$
b) $\frac{dn}{dt} = -k$
c) $\frac{dn}{dt} = k$
d) $\frac{dn}{dt} = -kn$

32. The asymptotes to the hyperbola $9x^2 - 4y^2 = 36$ are $y =$

a) $\pm\frac{9}{4}x$
b) $\pm\frac{4}{9}x$
c) $\pm\frac{2}{3}x$
d) $\pm\frac{3}{2}x$

33. If $5x \equiv 2 \pmod{16}$, then one value of x is

a) 0
b) 4
c) 6
d) 10

34. The integrating factor of the differential equation $\frac{dy}{dx} + y\tanh x = \cos x$ is

a) $\cosh x$
b) $\operatorname{sech} x$
c) $e^{\cosh x}$
d) $e^{\operatorname{sech} x}$

35. The negation of the statement $P \wedge \sim Q$ is

a) $P \vee Q$
b) $\sim P \wedge Q$
c) $P \wedge Q$
d) $\sim P \vee Q$

36. The work done against a force $\mathbf{F}$ in moving its point of action from point $\mathbf{a}$ to point $\mathbf{b}$ is

a) $(\mathbf{b} - \mathbf{a}) \cdot \mathbf{F}$
b) $(\mathbf{a} - \mathbf{b}) \cdot \mathbf{F}$
c) $(\mathbf{b} - \mathbf{a}) \times \mathbf{F}$
d) $(\mathbf{a} - \mathbf{b}) \times \mathbf{F}$

37. Given that $h\left(\frac{dy}{dx}\right)_n \approx y_{n+1} - 2y_n$ and $\frac{dy}{dx} = x + 2$, then $y_{n+1} \approx$

a) $2y_n + hx_n + 2$
b) $2y_n + h(x_n + 2)$
c) $2y_n + h(x_n - 2)$
d) $2y_n - h(x_n + 2)$

38. The function $f(x) = \dfrac{(x^2 + x)(x + 2)(x + 3)}{x(x - 1)(x - 2)}$ has a removable discontinuity at $x =$

a) 0
b) 1
c) 2
d) 3

39. A particle moves along the curve $r = 1 - \cos\theta$ with constant angular velocity ω. The maximum speed occurs for $\theta =$

a) $\dfrac{\pi}{2}$
b) π
c) $\dfrac{3\pi}{2}$
d) 2π

40. Which one of the following lines is invariant under the transformation with matrix $M = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}$?

a) $\dfrac{1}{2}x = y = \dfrac{1}{3}z$
b) $2x = y = 3z$
c) $\dfrac{1}{2}x = y = z$
d) $x = \dfrac{y}{2} = \dfrac{z}{3}$

41. For what value(s) of n is $\displaystyle\int_0^2 x^n\,dx = \int_0^2 (2 - x)^n\,dx$?

a) $n = 2k,\ k \in \mathbb{Z}$
b) $n = 2k + 1,\ k \in \mathbb{Z}$
c) $n \in \mathbb{R}$
d) $n \notin \mathbb{R}$

42. $z = \cos\dfrac{\pi}{6} + i\sin\dfrac{\pi}{6}$, then $\left|z + \dfrac{1}{z}\right| =$

a) $\dfrac{\sqrt{3}}{2}$
b) 1
c) $\sqrt{3}$
d) $2\sqrt{3}$

43. A binary operation $\bullet$ is defined on $\mathbb{R}$ as $x \bullet y = x + y + 1$. The inverse of 2 is

a) -4
b) -2
c) 0
d) 4

44. The first three terms in the Maclaurin expansion of e^{3x} are

a) $1 + 3x + \dfrac{9}{2}x^2$
b) $1 + 3x + 9x^2$
c) $1 + 3x + \dfrac{9}{4}x^2$
d) $1 + 3x + 3x^2$

45. A sequence (u_n) is defined recursively by $u_0 = 3$, $u_{n+1} = \dfrac{2}{3}u_n$. Then $u_n =$

a) $2^n\,3^{1-n}$
b) $2^n\,3^n$
c) $2^{1-n}\,3^n$
d) $2^n\,3^{n-1}$

46. The asymptote to the curve $y + 1 = 2^x$ is

a) $x = 0$
b) $y = -1$
c) $y = 1$
d) $x = 1$

47. The arc length of the curve $f(x) = \cosh x$ for $0 \le x \le \ln 2$ is

a) $\dfrac{3}{4}$
b) $\dfrac{5}{4}$
c) $\dfrac{5}{2}$
d) $\dfrac{3}{2}$

48. For any three non-singular matrices A, B and C, $A^{-1}(BC)^{-1} =$

a) $A^{-1}C^{-1}B^{-1}$
b) $C^{-1}B^{-1}A^{-1}$
c) $A^{-1}B^{-1}C^{-1}$
d) $B^{-1}A^{-1}C^{-1}$

49. A force $\mathbf{F} = (3\mathbf{i} + \mathbf{j} - 2\mathbf{k})\ \mathrm{N}$ acts through the point $A(2, 2, 2)$. The moment of $\mathbf{F}$ about the point $B(0, 3, 1)$ is

Options as printed: A) $3\mathbf{i} - 7\mathbf{j} - 5\mathbf{k}$ B) $-3\mathbf{i} + 7\mathbf{j} - 5\mathbf{k}$ C) $-3\mathbf{i} - 7\mathbf{j} + 5\mathbf{k}$ D) $-3\mathbf{i} - 7\mathbf{j} - 5\mathbf{k}$

50. Which one of the following functions is NOT differentiable in the interval $0 \leq x \leq 2$?

a) $y = |x - 1|$
b) $y = x^2$
c) $y = -x^2$
d) $y = |x - 2|$

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

View the correction

Scan me

edukora.africa - Free-access paper, feel free to share. The correction is for subscribers only.